L10. Research Ethics

Morality

Descriptive view of morality

What it is?

The descriptive view of morality says that morality is simply the rules and codes of conduct that people actually accept. This view is problematic because many accepted rules are not moral, and it leads to relativism where no action can be judged as truly morally wrong. Another way to define it as a code of conduct that is put forward by a society, group, or individual, that is accepted as a guide to behaviour

Example(s):

- Driving on the right side of the road is a shared rule, but breaking it is not immoral, only a mistake.
- Etiquette rules, like not speaking with your mouth full, are not moral rules.
- If morality were only based on accepted codes, even clearly unethical actions could not be condemned if a group accepted them.

Normative View

What it is?

The **normative view of morality** says that morality is not based on what people actually do or believe. Instead, it is based on what **all rational people would agree is right or wrong if they were thinking clearly and fairly**. These are ideal conditions where people do not know their own interests, power, or social position. This makes morality objective rather than relative to cultures or groups. It also allows us to criticise laws, traditions, or scientific practices even if many people accept them.

Example(s):

- **Harming others:** Rational people, thinking impartially, would agree that harming innocent people is wrong, no matter where or when it happens.
- **Scientific experiments:** Experiments that seriously harm participants are morally wrong even if scientists or governments approve of them.
- **Law vs morality:** A law can exist and be followed, but still be immoral if rational people would reject it under fair conditions.
- **Research ethics:** Rules like informed consent and avoiding unnecessary harm are moral rules because rational people would support them, not just because they are written in guidelines.

Ethics

What it is?

Ethics is a theory that explains **which rules of behaviour we should follow and why**. It gives clear, rational reasons for judging actions as right or wrong. Ethics is not about describing what people do, but about **justifying what people ought to do**.

Example(s):

- **Research ethics:** Explains why harming research participants is wrong and why informed consent is required.
- **Honesty:** Gives reasons why lying is wrong, even if it is common or useful.
- **Engineering practice:** Justifies why safety should matter more than saving money.

Moral vs Ethics

What it is?

The **moral vs ethical** distinction says that **morals** are about whether an action, intention, or decision is **good or bad**, while **ethics** is about **explaining and justifying why** it is good or bad. Morals describe the moral status of actions. Ethics gives the reasons and theories behind that judgment. In everyday language, this difference is often not kept, and the words are used as if they mean the same thing.

Example(s):

- **Morals:** Lying to a friend is morally wrong.
- **Ethics:** Ethics explains why lying is wrong, for example because it breaks trust or harms others.
- **Everyday language:** The same action might be called *immoral* or *unethical* with no real difference in meaning.

Three Frameworks of Normative Ethics

Deontology

What it is?

Deontology is an ethical view that says an action is right or allowed if it **follows certain duties or rules**. These duties come from roles we have or from moral obligations, such as **respecting the rights of other people**. The focus is on what we must do, not on the outcomes.

Example(s):

- **Respecting rights:** It is wrong to harm someone because we have a duty to respect their rights.
- **Telling the truth:** Lying is wrong because honesty is a duty, even if lying could lead to good results.
- **Professional roles:** A doctor must respect patient autonomy because it is part of their duty.

Consequentialism

What it is?

Consequentialism is an ethical view that says an action is right or wrong **only because of its consequences**. The action itself does not

matter, only the results it leads to. A common form is **utilitarianism**, which aims to produce the most happiness or overall benefit.

Example(s):

- **Saving many people:** An action is right if it saves more lives than the alternatives.
- **Policy decisions:** A law is good if it increases overall well-being, even if some individuals are harmed.
- **Lying:** Lying can be acceptable if it leads to better outcomes than telling the truth.

Virtue ethics

What it is?

Virtue ethics is an ethical view that focuses on **character rather than rules or outcomes**. An action is right if it shows good character traits, called virtues, such as honesty, courage, humility, and respect. The goal is to become a good person and to develop human abilities in a balanced way.

Example(s):

- **Honesty:** Telling the truth because honesty is a virtue of a good person.
- **Courage:** Acting bravely in a difficult situation, even when it is risky.
- **Professional life:** A scientist acting carefully and responsibly because integrity is part of good character.

Categorical Imperative

What it is?

The **categorical imperative** is a rule in **Kant's ethics**. It says that an action is right only if it follows a **principle that everyone could reasonably accept and follow**. The action must be acceptable as a general rule, not just useful in one case.

Example(s):

- **Lying:** Lying is wrong because if everyone lied, trust would break down, so the principle cannot be defended for everyone.

- **Keeping promises:** Keeping promises is right because it can be accepted as a rule for all people.
- **Respecting people:** Using others only as tools is wrong because no one could accept that as a general principle.

Informed Consent

What it is?

Informed consent means that a person agrees to take part in an activity, such as research, **in a proper and ethical way**. This requires three things: the person is given enough information, joins by free choice, and is able to understand and decide.

Example(s):

- **Informed:** A research participant is told the purpose of the study, what will happen, and possible risks.
- **Voluntary:** The participant joins without pressure, threats, or rewards that force the decision.
- **Decision-capacitated:** The participant is mentally able to understand the information and make a clear decision.

Scientific Misconduct

Fabrication

What it is?

Fabrication is a form of scientific misconduct where a researcher **intentionally invents data or results**. The information is not based on real experiments or observations and is created to **mislead others**.

Example(s):

- **Fake data:** Reporting measurements that were never actually collected.
- **Invented results:** Claiming that an experiment produced certain outcomes when the experiment was never done.
- **Misleading publications:** Publishing graphs or tables based on made-up numbers.

Falsification

What it is?

Falsification is a form of scientific misconduct where a researcher **intentionally changes, manipulates, or removes data or results** without scientific justification. The goal is to **mislead the reader** and make the results look better or more convincing than they really are.

Example(s):

- **Changing data:** Altering measurements so they better fit a hypothesis.
- **Removing results:** Leaving out data points that contradict the expected outcome.
- **Manipulating images:** Editing images or graphs to exaggerate effects or hide problems.

Plagiarism

What it is?

Plagiarism is a form of academic misconduct where someone **uses another person's ideas, methods, results, or words as if they were their own**, without proper acknowledgement. It misleads others about who did the original work.

Example(s):

- **Copying text:** Using sentences from a paper or book without citing the source.
- **Stealing ideas:** Presenting another researcher's idea or argument without credit.

- **Uncredited results:** Using data or findings from someone else's study without mentioning them.

Whistleblowing

What it is?

Whistleblowing means **reporting illegal or unethical actions** that happen inside an organisation. It is usually done by someone who works in or is closely connected to the organisation, with the aim of stopping harm or wrongdoing.

Example(s):

- **Research misconduct:** A scientist reports fabricated or falsified data in their lab.
- **Safety issues:** An engineer reports that a company is hiding serious safety risks.
- **Corruption:** An employee reports fraud or misuse of public funds.

Authorship and some variants

Author

What it is?

An **author** is someone who has made a **major and responsible contribution** to a scientific or academic work. To be an author, the person must contribute to the idea, help write the work, approve the final version, and take responsibility for the whole content.

Example(s):

- **Research design:** A researcher helps plan the study and its main ideas.

- **Writing:** A contributor helps draft or revise the article in a meaningful way.
- **Final approval:** The person reads and approves the final version before submission.
- **Responsibility:** The author accepts responsibility for the accuracy and integrity of the entire work.

Gift Authorship

What it is?

Gift authorship means listing someone as an author **even though they did not make a real contribution** to the work. This is often done for social reasons, such as politeness, hierarchy, or gaining favour, and it is considered unethical.

Example(s):

- **Senior pressure:** Adding a supervisor as an author even though they did not help with the research or writing.
- **Favour exchange:** Listing a colleague as an author to return a favour.
- **Status boosting:** Including a well-known name to make the paper seem more important.

Ghost authorship

What it is?

A **ghost author** is a person who **made a substantial contribution to a work but is not listed as an author**. This hides who actually did the work and is considered unethical.

Example(s):

- **Uncredited writer:** A researcher writes large parts of a paper but is not named as an author.
- **Industry influence:** A company employee helps write a scientific article, but only academic researchers are listed.

- **Student work:** A student does most of the research and writing, but only a supervisor is named as author.

Frivolous Research

What it is?

Frivolous research is research done **mainly for entertainment or curiosity**, without a serious scientific, practical, or social purpose. As far as one can tell, the goal is not to gain useful knowledge or solve an important problem.

Example(s):

- **Playful experiments:** Studying a topic only because it is amusing, with no clear research value.
- **Pointless measurements:** Collecting data that does not help answer any meaningful question.
- **Novelty projects:** Doing research mainly to be funny or unusual, not to contribute to science or society.

Effective altruism

What it is?

Effective altruism is a way of thinking and a movement that aims to **help others as much as possible**. It focuses on using **evidence and careful reasoning** to choose actions that do the most good with the resources available.

Example(s):

- **Charity choice:** Donating money to causes that save or improve the most lives per krona or dollar.
- **Career choice:** Choosing a job where you can have the biggest positive impact on others.
- **Global health:** Supporting programs like vaccinations because they are highly effective at reducing suffering.